Exploding Arrow

You are given three integers N, M, and K. You have K arrows, and your goal is to destroy N targets. Each target i has an initial hit points (hp) value of a_i. To destroy a target, its hp must be less than or equal to 0.

When you shoot an arrow with power X at target i, it decreases the hp of all targets with index $j \geq i$ by an amount calculated as follows:

$$\text{Damage to target } j = \max(0, M \cdot X - (j-i)^2)$$

Your task is to find the minimum possible value of X such that you can destroy all targets, given that you can choose the indices of your arrows optimally.

Input Format

- The first line contains three integers N, M, and K: the number of targets, the multiplier for the arrow power, and the number of arrows, respectively.
- The second line contains N integers $a_0, a_1, \ldots, a_{N-1}$ representing the initial hp of the targets.

Output Format

- Output a single integer, the minimum value of X such that you can destroy all targets.

Constraints

Denote $A = a_0 + a_1 + \ldots + a_{N-1}$. In all test cases we have - $1 \leq M \leq 10^9$. - $1 \leq N \leq 2 \times 10^5$. - $1 \leq K \leq 10^9$. - $1 \leq a_i \leq 10^9$.

Constraints for each test case: 1. Sample 2. $K = 1$. 3. $A \leq 10000 \cdot MK$ and $N \leq 100$. 4. $A \leq 5 \times 10^6 \cdot MK$ and $N \leq 5000$. 5. $N \leq 1000$. 6. $N \leq 1000$. 7. $N \leq 2 \times 10^5$. 8. $N \leq 2 \times 10^5$.

Example

Input

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4 1 2
3 3 2 4
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Output

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5
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Explanation

In the example, you can destroy all targets with the following strategy:

1. Shoot the first arrow (index 0) with power $X = 5$:
   - Damage to target 0: $1 \cdot 5 - (0-0)^2 = 5 \rightarrow a[0] = 3 - 5 = -2$
   - Damage to target 1: $1 \cdot 5 - (1-0)^2 = 4 \rightarrow a[1] = 3 - 4 = -1$
   - Damage to target 2: $1 \cdot 5 - (2-0)^2 = 1 \rightarrow a[2] = 2 - 1 = 1$
   - Damage to target 3: $1 \cdot 5 - (3-0)^2 = -4 \rightarrow a[3] = 4 - 0 = 4$
2. Shoot the second arrow (index 2) with power $X = 5$:
   - Damage to target 2: $1 \cdot 5 - (2-2)^2 = 5 \rightarrow a[2] = 1 - 5 = -4$
   - Damage to target 3: $1 \cdot 5 - (3-2)^2 = 4 \rightarrow a[3] = 4 - 4 = 0$

All targets are eliminated.